# Exam #1 (Sta-209, S25)

Ryan Miller

**Directions**

- Answer each question using *no more than specified number of sentences* and not attempt to avoid these guidelines by using run-on sentences. Answers that are unnecessarily verbose may result in point loss.
- Do not include superfluous information in your answers, you may be penalized if you make an inaccurate statement even if you go on to provide a correct answer. Your answers should be clear, concise, and include only what is needed to answer the question that was asked.

**Formula Sheet**

**Definitions**:

- **Risk**: relative frequency of an event/outcome
- **Relative Risk**: ratio of the risks across two groups
- **Odds**: ratio of how often an event/outcome is observed relative to how often it is not observed
- **Odds ratio**: ratio of odds across two groups

**Formulas**:

Mean:

$$\bar{x} = \tfrac{1}{n}\sum_{i=1}^{n} x_i$$

Standard deviation

$$s = \sqrt{\tfrac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Pearson's Correlation Coefficient:

$$r = \tfrac{1}{n-1}\sum_{i=1}^{n}\big(\tfrac{x_i-\bar{x}}{s_x}\big)\big(\tfrac{y_i-\bar{y}}{s_y}\big)$$

Simple linear regression (theoretical model):

$$Y = b_0 + b_1X + \epsilon$$

Simple linear regression (fitted model):

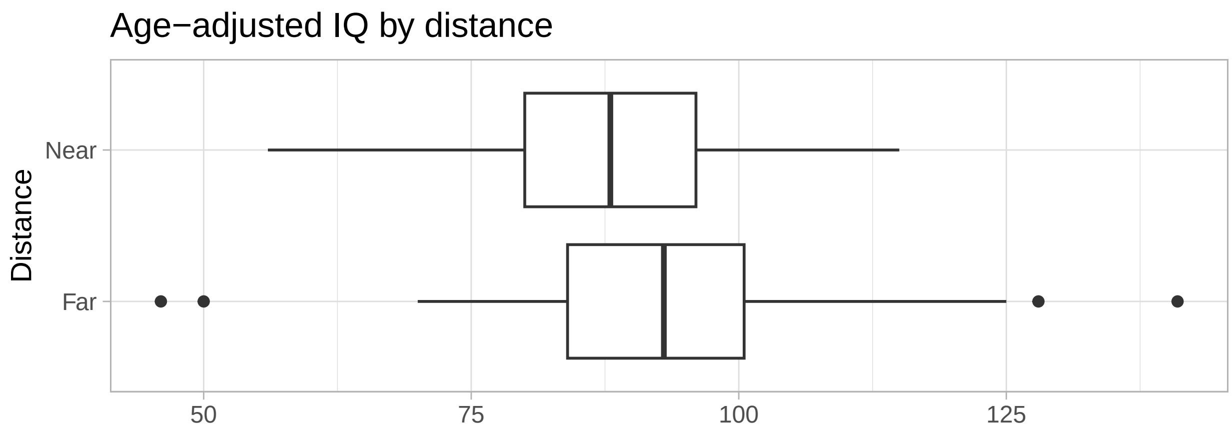
$$\hat{y} = \hat{b}_0 + \hat{b}_1x$$

Coefficient of Determination ($R^2$):

$$R^2 = \frac{\sum_{i=1}^{n}(\hat{y}_i - \overline{y})^2}{\sum_{i=1}^{n}(y_i - \overline{y})^2}$$

**Question #1**

As part of a study done by the Center for Disease Control (CDC), researchers collected data on two groups of children aged 3-15 living in El Paso Texas. The "Near" group consisted of 57 children living within 1-mile from a local lead smelter, while the "Far" group consisted of 67 children living at least 1-mile from the smelter. The researchers recorded the age-adjusted IQ score of each child as the study's outcome variable. The figure below displays the results:




**Part A**: If the data used to create the figure shown above were stored in "tidy" format with each case as a row and each column as a variable, how many rows and columns would this data frame contain? Do not consider any subject identifiers or variables not shown in the figure. You do not need to explain your answer.

**Part B**: Consider the distribution of age-adjusted IQ scores in the "Near" group. Would you expect the *mean* age-adjusted IQ of this group to be *approximately equal* or *substantially different* from the *median* age-adjusted IQ score of the group? Briefly explain your answer using at most 2-sentences.

**Part C**: Based upon the previous figure, do you believe there is an association between a child's age-adjusted IQ and whether they live near or far from the lead smelter? Briefly explain your answer using at most 2-sentences.

**Question #2**

A research study published in the British Medical Journal (BMJ) looked at two treatments, "open surgery" or "PN" (an abbreviation of Percutaneous Nephrolithotomy), for a sample of 550 patients suffering from kidney stones. For each patient, a binary outcome of "success" or "failure" of the treatment was recorded. The table below summarizes the distribution of outcomes for each treatment:

| | Success | Failure | Total |
|---|---|---|---|
| Open Surgery | 156 | 44 | 200 |
| PN | 289 | 61 | 350 |
| Total | 445 | 105 | 550 |

**Part A**: There are two variables involved in the previously given table. Which of these is the *explanatory variable* and which is the *response variable*? Briefly explain your answer using at most 2-sentences.

**Part B**: Among the patients who were successfully treated, what proportion of these patients received the PN treatment? Show your calculation. It is acceptable to leave your answer as a fraction.

**Part C**: Your friend looks at the proportion you calculated in Part B and concludes that PN treatment is more effective than Open Surgery since the proportion is larger than 0.5. Do you agree with your friend's interpretation of this proportion? Briefly explain your answer using at most 3-sentences.

**Part D**: Calculate the *odds of success* for each treatment. Show your calculations and be sure to clearly label your results.

**Part E**: Calculate the odds ratio that compares the odds of success among patients who received PN relative to patients who received Open Surgery. Provide a 1-sentence interpretation of this odds ratio.

**Part F**: The researchers in this study collected another variable, whether the patient's kidney stones were "large" (mean diameter $\geq$ 2cm) or "small (mean diameter $<$ 2cm). The table below summarizes the success rate (number of successes divided by number of cases) for each combination of treatment and stone size. Using this table, which treatment had a higher *overall success rate*, regardless of stone size (ie: a higher marginal effect)? You do not need to explain your answer.

| | Open Surgery | PN |
|---|---|---|
| Small Stones | 46/50 (92.0%) | 234/270 (86.7%) |
| Large Stones | 110/150 (73.3%) | 55/80 (68.9%) |
| Total | 156/200 (78.0%) | 289/350 (82.6%) |

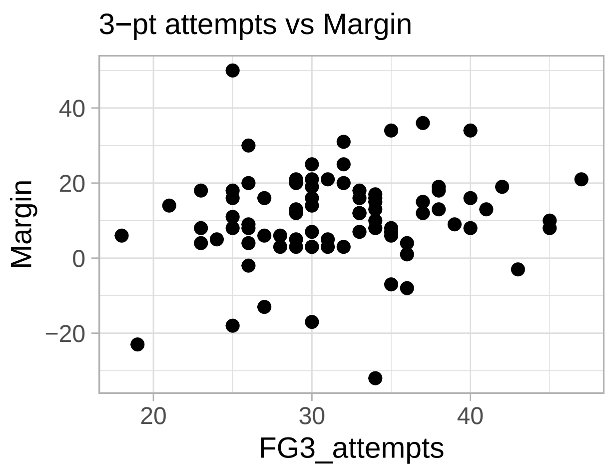
**Part G**: How is it possible that the marginal effect suggests one treatment is more effective, but when the data are stratified the conditional effects in every stratum suggest the other treatment is more effective? Explain your answer in no more than 4-sentences.

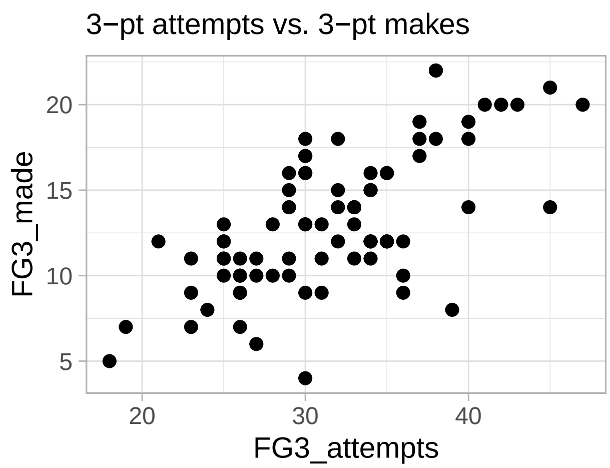
**Question #3**

This question uses data from the Golden State Warrior's 2015-16 season where the team set a record for the most regular season wins in NBA history. Each row in this data set describes one of the season's 82 games. Four variables are involved in this question:

- `Margin` - The margin of victory (if positive) or loss (if negative) for the Warriors. This is calculated by subtracting the points scored by the Warrior's opponent from the points scored by the Warriors.
- `FG3_attempts` - The number of three-point shots *attempted* by the Warriors in the game.
- `FG3_made` - The number of three-point shots *made* by the Warriors in the game.
- `Location` - Whether the game was played on the Warrior's home court or away.

Shown below are two scatter plots involving various combinations of these variables:




**Part A**: Briefly describe the relationship between `FG3_attempts` and `FG3_made`. Be sure to address all relevant aspects of this relationship.

**Part B**: Shown below is a *fitted simple linear regression model* that uses `FG3_attempts` as the explanatory variable and `Margin` as the response variable. Briefly interpret the estimated coefficient of the variable `FG3_attempts` in this model.

```
## Part B Model
part_b_model = lm(Margin ~ FG3_attempts, data = gsw)
coef(part_b_model)
```

```
##  (Intercept) FG3_attempts
##    1.7459665    0.2850427
```

**Part C**: Shown below is a *fitted multivariable linear regression model* that uses `FG3_attempts` and `FG3_made` as explanatory variables and `Margin` as the response variable. Explain why the estimated coefficient of the variable `FG3_attempts` is negative in this model, but positive in the model from Part B. You should limit your explanation to around 3 sentences.

```
## Part C Model
part_c_model = lm(Margin ~ FG3_attempts + FG3_made, data = gsw)
coef(part_c_model)
```

```
##  (Intercept) FG3_attempts     FG3_made
##    3.3479640   -0.4529117    1.6540520
```

**Part D**: Shown below is a third *fitted multivariable linear regression model* that uses three explanatory variables: `Location`, `FG3_attempts`, and `FG3_made`, and `Margin` as the response variable. Briefly interpret *the intercept* of this model.

```
## Part D Model
part_d_model = lm(Margin ~ Location + FG3_attempts + FG3_made, data = gsw)
coef(part_d_model)
```

```
##  (Intercept) LocationHome FG3_attempts     FG3_made
##   -0.8398691    7.4229580   -0.4413523    1.6625005
```

**Part E**: Interpret the coefficient of the variable `LocationHome` in the model from Part D.

**Part F**: For each of the following combinations of variables, state whether the pair is "associated" or "nearly independent". You should only consider marginal associations, and you are not required to provide a justification, but you may do so for a chance at partial credit. You should consider all information provided in previous parts of Question 3.

1. `Location` and `Margin`

2. `Location` and `FG3_made`

3. `FG_attempts` and `FG3_made`